Reg. No. | | | | | | | | | | | [MSCCH406]

FOURTH SEMESTER UNIVERSITY EXAMINATION, JUNE 2016

M.Sc. (CHEMISTRY)

PHARMACEUTICAL AND INDUSTRIAL CHEMISTRY

Time : 03 Hours **Maximum Marks:80**

Instructions: 1. Questions 1 to 6 carry 15 marks each, answer any four.

2. Question 7 is compulsory, carries 20 marks.

1. a. Discuss the various factors governing drug-design.

b. Enumerate the different cardinal objectives of variation methods by giving appropriate examples.

c. Differentiate the basic concepts of 'analogues' and 'prodrugs' with suitable examples of parent drug molecule(s). **5+5+5**

2. a. Explain the synthesis of acyclovir.

b. Write notes on the synthesis of sorbitrate and methyldopa.

c. What are antimalarial drugs? Outline the synthesis and SAR of chloroquine. **4+5+6**

3. a. Explain the synthesis and mode of action of DDT and heptachlor.

b. What are pyrethrins and pyrethroids? Give the structures of any two natural pyrethroids.

c. Write notes on malathion and diazenon. **6+3+6**

4. a. Write a note on the classification of dyes based on their constitution.

b. What are paints? Explain their properties. **10+5**

5. a. Explain the free radical mechanism involved in styrene polymerization.

b. What are thermosetting and thermoplastic polymers? Explain with suitable examples.

c. Write a brief note on epoxy resin. **5+5+5**

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6. a. Define cracking. Explain fixed bed catalytic cracking.

b. Describe any two methods for the stabilization of gasoline.

c. Write a note on 'knocking' and 'octane number'. **5+5+5**

7. **Answer any *TEN* of the following:** **10×2=20**

a. Differentiate pharmacodynamics and pharmacokinetics.

b. Define 'computer-aided drug design'.

c. Explain the synthesis of sulphonamide.

d. What are analgesic and antipyretic drugs? Write the structures of any two analgesic-antipyretics drug.

e. Define pesticides and insecticides.

f. Write the structures of sevin, and beygon.

g. Define dyes. Explain any two of their properties.

h. What is thinner? Explain.

i. Write a note on urea-formaldehyde and its applications.

j. Differentiate weight average molecular weight from number average molecular weight of polymers.

k. Describe the advantages of compressed natural gas (CNG).

l. Define blending and doping of anti-knocking agents.